This demo contains:

- **Selecting Columns and Filtering Rows**
- **Creating a Column with an Expression**
- **Summarizing and Filtering by Groups**
- **Joining Tables**

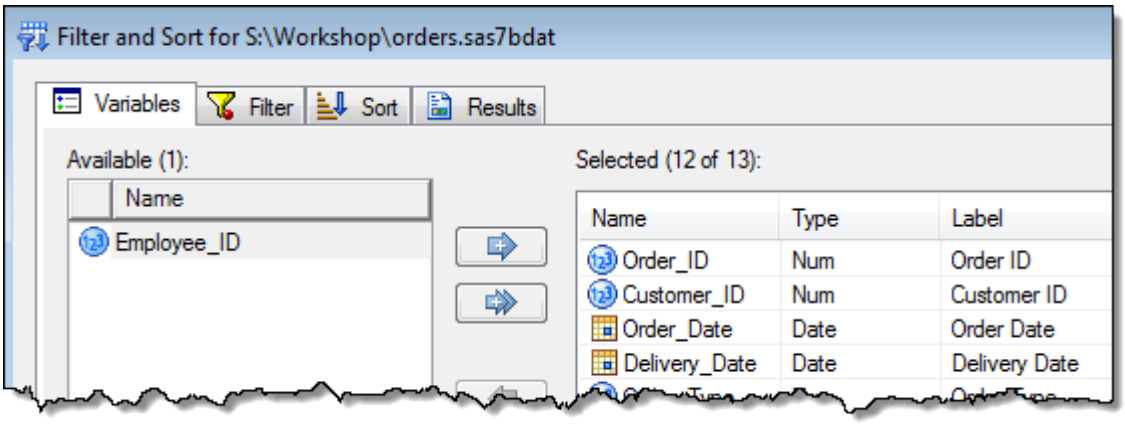
Selecting Columns and Filtering Rows

Orion Star would like to analyze Internet sales that occurred since the beginning of 2013. To prepare the data for input to the various analytic tasks, the company must generate a new subset from the **orders** table. This query can be created with either the Filter and Sort task or the Query Builder.

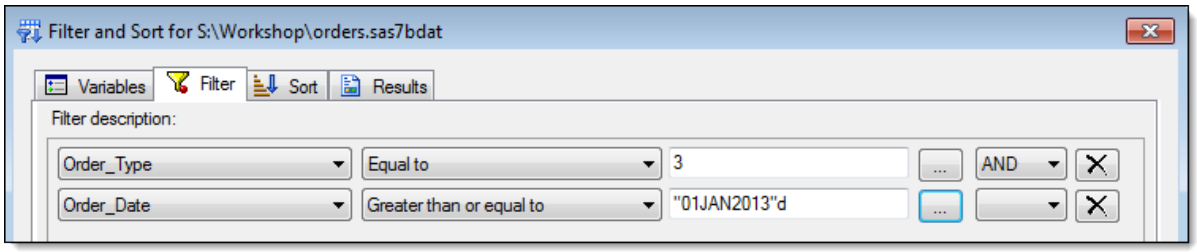
1. Create a new process flow and rename it as **Creating Simple Queries**.
2. Add the **orders** SAS data set.
3. To create the new data set that includes Internet orders placed on or after 01JAN2013, begin by clicking the **Filter and Sort** tab on the data grid.

   You can also right-click **orders** in the project tree or process flow, and select **Filter and Sort**.

4. On the Variables tab, click to add all columns. All Internet orders have the **Employee_ID** column equal to **99999999**, so it can be removed from the result. Highlight **Employee_ID** in the Selected pane and click .



5. Click the **Filter** tab. Define the two filters shown in the display below.



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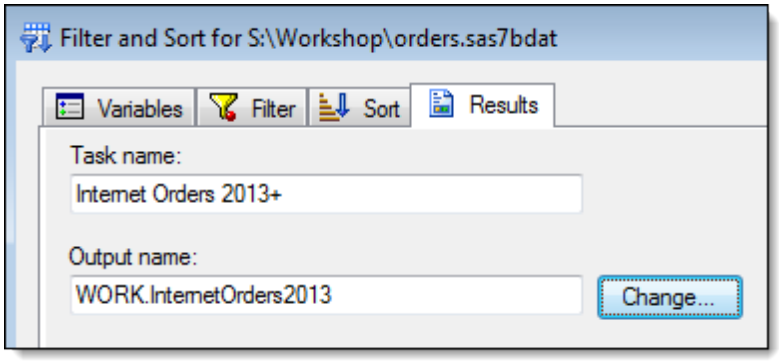
Order_Type is a coded column where 1 represents retail sales, 2 represents catalog sales, and 3 represents Internet sales.

SAS dates are stored as the number of days since 01JAN1960. To build a filter based on a specific date, you can either select a particular date from the data by clicking ... or you can enter a SAS date constant. A SAS date constant must be entered in this form: **"DDMONYYYY"d**. For example, January 1, 2013 would be entered as **"01JAN2013"d**.

6. Click the **Sort** tab. Select **Order_Date** as the sort variable and change the sort sequence to **Descending**.

7. Click the **Results** tab. Enter **Internet Orders 2013+** in the **Task name** field. Click **Change** and enter **internetorders2013** in the **File name** field. Select **Save**.



8. Click **OK** and verify the results. This data source is a SAS table that can be exported or used as input for other tasks.

   Partial Results (8 of 55 rows, 5 of 12 columns)

	Order_ID	Customer_ID	Order_Date	Delivery_Date	Order_Type	
1	1244086685	14104	03DEC2014	06DEC2014	3	2
2	1244086685	14104	03DEC2014	06DEC2014	3	2
3	1243644877	70079	16OCT2014	19OCT2014	3	2
4	1243485097	11	2[illegible]SEP2014	02OCT2014	3	2
5	1243462945	24	25SEP2014	28SEP2014	3	2
6	1243279343	27	05SEP2014	10SEP2014	3	2
7	1243165497	70201	24AUG2014	29AUG2014	3	2
8	1243165497	70201	24AUG2014	29AUG2014	3	2

9. The Query Builder can also be used to produce a similar result. To compare the two tasks, double-click the **orders** table in the project. Select **Query Builder** from the data grid.

   You can also right-click **orders** in the project and select **Query Builder**.

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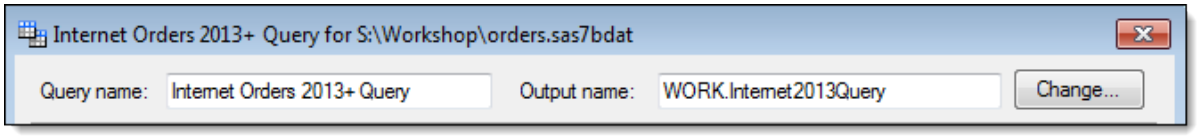
10. The Query Builder enables you to name the query icon and the name and storage location of the SAS table that you created. Enter **Internet Orders 2013+ Query** in the **Query name** field. Click **Change** next to the **Output name** field. Enter **internet2013query** in the **File name** field and select **Save**.

    The table is saved to the default SAS library. However, a different library can be selected if it is available.

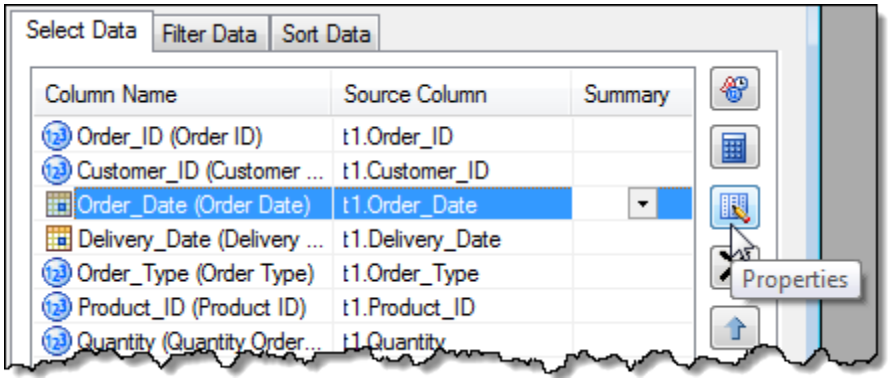


11. To select all columns, select **t1 (orders)** in the Selection pane and drag and drop it onto the Select Data tab. All columns are added to the query. Remove the **Employee_ID** column by highlighting it on the Select Data tab and clicking ✕.

    Columns can be added individually either by double-clicking or by dragging and dropping. Multiple columns can be selected by holding down the Ctrl key. You can use the Shift key to select a range of columns.

12. Column properties can also be viewed or modified on the Select Data tab. Highlight **Order_Date** and click to view the Properties window.

    You can also double-click on a column to access the Properties window.



13. To apply a format to this column, select **Change**. In the Formats window, select **Date** from the Categories pane and **MMDDYYw.d** from the Formats pane. Change the overall width to **10**. Click **OK** ⇨ **OK** to return to the Query Builder.

14. To include only Internet orders placed on or after 01JAN2013, click the **Filter Data** tab. Drag and drop **Order_Type** to the Filter Data tab to start the Basic Filter Wizard.

    You can also click and select **Basic Filter**.

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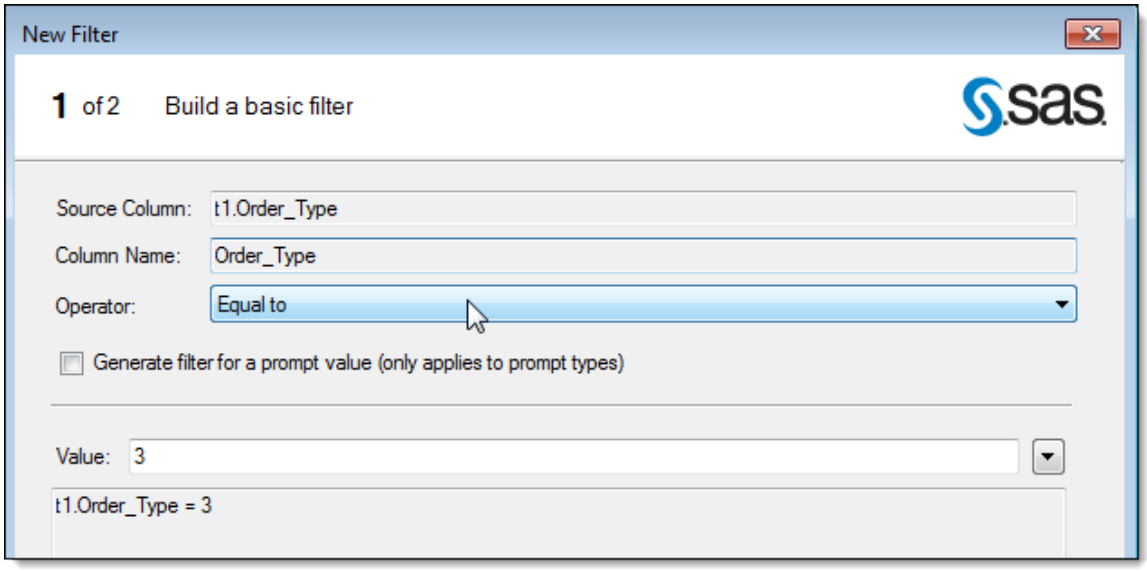
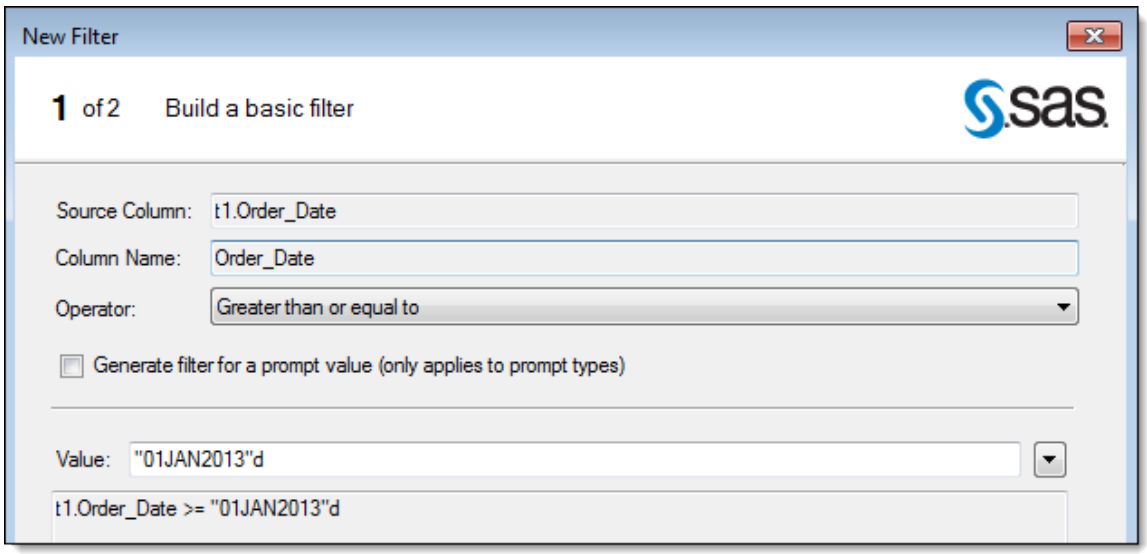
15. In Step 1, change the operator to **Equal to** and enter **3** in the **Value** field.



16. Click **Next** to verify the filter and click **Finish**.

17. Drag **Order_Date** to the Filter Data tab. In Step 1, change the operator to **Greater than or equal to** and enter **"01JAN2013"d** in the **Value** field.



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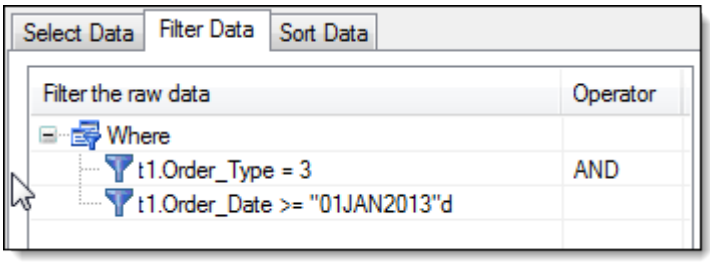
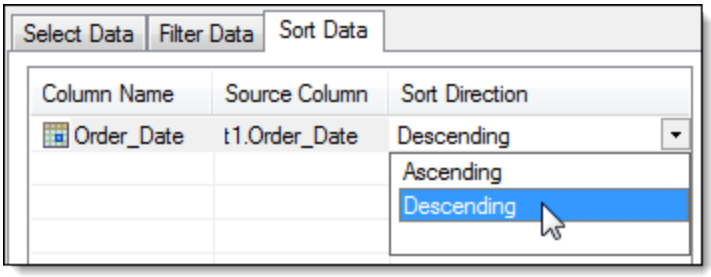
18. Click **Next** to verify the filter and click **Finish**.

    The two filters are automatically connected by an AND operator.



19. Click the **Sort Data** tab. Drag and drop **Order_Date** onto the Sort Data tab and change the sort direction to **Descending**.



20. Click **Run** to execute the query and view the resulting SAS table.

    Partial Results (8 of 55 rows, 6 of 12 columns)

	Order_ID	Customer_ID	Order_Date	Delivery_Date	Order_Type	P
1	1244086685	14104	12/03/2014	06DEC2014	3	2
2	1244086685	14104	12/03/2014	06DEC2014	3	2
3	1243644877	70079	10/16/2014	19OCT2014	3	2
4	1243485097	11	09/28/2014	02OCT2014	3	2
5	1243462945	24	09/25/2014	28SEP2014	3	2
6	1243279343	27	09/05/2014	10SEP2014	3	2
7	1243165497	70201	08/24/2014	29AUG2014	3	2
8	1243165497	70201	08/24/2014	29AUG2014	3	2

21. Save the project.

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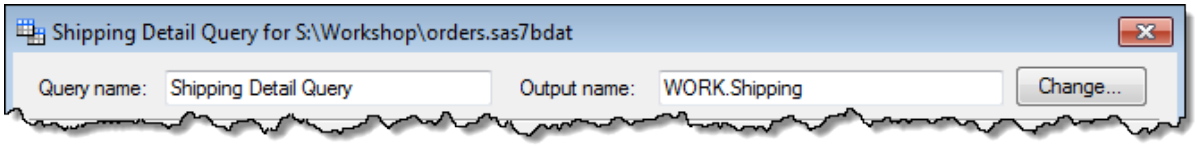
Creating a Column with an Expression

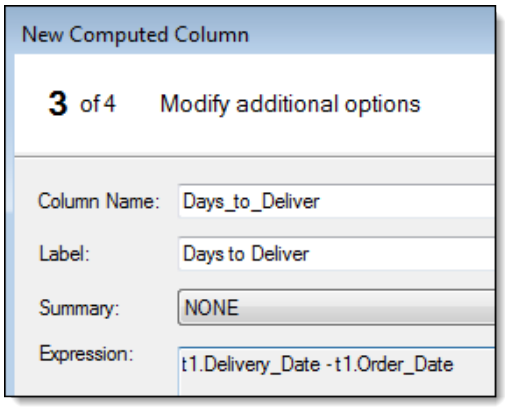
Orion Star would like to analyze shipment methods by determining how many days passed between each order date and delivery date. The company also wants to calculate the total amount invoiced to the customer, which is the sum of the total retail price and shipping charges. Use the Query Builder to create a table named **shipping** with the new computed columns.

1. In the process flow, double-click the **orders** table. Select **Query Builder**.
2. Enter **Shipping Detail Query** in the **Query name** field. Click **Change** and enter **shipping** in the **File name** field. Select **Save**.



3. Verify that the **Select Data** tab is active. Add the following columns from the **orders** table: **Order_ID**, **Order_Date**, **Delivery_Date**, **Product_ID**, **Total_Retail_Price**, **Shipping**, and **Profit**.
4. To add a new column to the query, begin by clicking on the Select Data tab, or you can select **Computed Columns** ⇨ **New**.
5. In Step 1, select **Advanced expression** ⇨ **Next**.
6. In Step 2, expand **Selected Columns**. Double-click **Delivery_Date** to add the column to the expression. Select or enter a minus sign. Double-click **Order_Date** to complete the expression. Click **Next**.

7. In Step 3, enter **Days_to_Deliver** in the **Column Name** field. Enter **Days to Deliver** in the **Label** field. Click **Next**.



8. In Step 4, review the summary of the new column's properties and click **Finish**.
9. Repeat the process to create the **Invoice_Amt** column. Begin by clicking .
10. In Step 1, select **Advanced expression** ⇨ **Next**.

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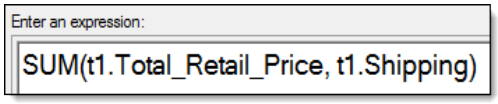
11. In Step 2, scroll down. Find the SUM function under the Functions folder. Double-click **SUM Function** to add it to the expression. Notice that the syntax for the SUM function is to the right of the list of functions.

12. Select **Favorites** ⇨ **Tables** to quickly collapse the Functions folder. Expand **Selected Columns**. Double-click **Total_Retail_Price**, and then enter or select the comma. Double-click **Shipping** to complete the expression as follows:



13. Click **Next**. In Step 3, enter **Invoice_Amt** in the **Column Name** field. Enter **Invoice Amount** in the **Label** field. To display the data values as currency, click **Change** next to the **Format** field. In the Format window, select **Currency** in the Categories pane and **DOLLARw.d** in the Formats pane. Change the overall width to **10** and the decimal places to **2**. Click **OK** ⇨ **Next**.

14. Verify the summary of the properties and click **Finish**. Both new columns are added to the Selection pane and the Select Data tab.

15. To emphasize the orders with the longest delivery time, sort the table in descending sequence by **Days_to_Deliver**. Click the **Sort Data** tab, and drag and drop the **Days_to_Deliver** column onto the tab area. Change the sort direction to **Descending**.

16. Run the query and examine the results. Save the project.

    Partial Results (8 of 617 rows)

	Order_ID	Order_Date	Delivery_Date	Product_ID	Total_Retail_Price	Shipping	Profit	Days_to_Deliver	Invoice_Amt
1	1236483576	22JUL2012	02AUG2012	240200200071	$74.80	$10.95	$34.80	11	$85.75
2	1237165927	27SEP2012	08OCT2012	240500100057	$76.80	$10.95	$42.60	11	$87.75
3	1236965430	08SEP2012	18SEP2012	240200100050	$27.00	$6.95	$13.70	10	$33.95
4	1243227745	31AUG2014	10SEP2014	230100500004	$6.40	$6.95	$3.70	10	$13.35
5	1241298131	29JAN2014	08FEB2014	240100400058	$37.40	$8.95	$7.75	10	$46.35
6	1231305521	27AUG2010	04SEP2010	220200100035	$125.20	$15.95	$62.40	8	$141.15
7	1232648239	07APR2011	15APR2011	210201000126	$6.50	$6.95	$4.20	8	$13.45
8	1233998114	27OCT2011	03NOV2011	220100400022	$98.90	$12.95	$51.20	7	$111.85

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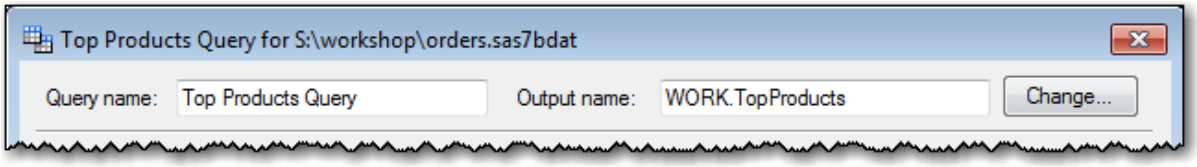
Summarizing and Filtering by Groups

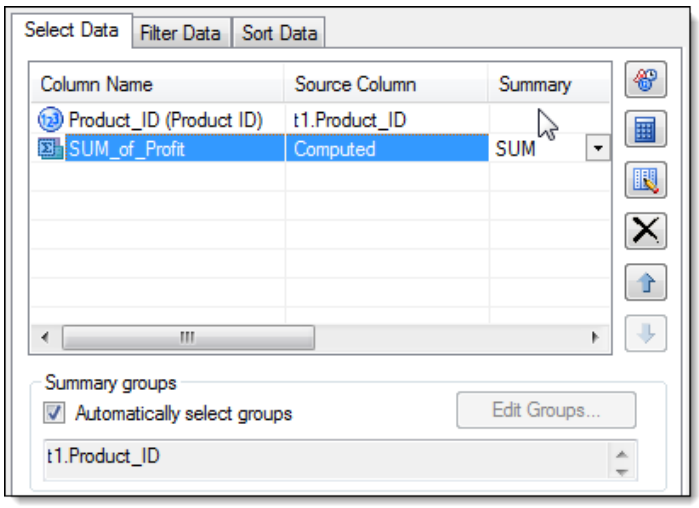
Orion Star would like to offer a sales promotion that highlights the most frequently purchased products. The company wants a list of all products with a total profit exceeding $500. Use the Query Builder to group, summarize, and filter the **orders** data set.

1. In the process flow, right-click the **orders** table in the project tree or process flow and select **Query Builder**.
2. Name the query **Top Products Query** and name the output table **TopProducts**.



3. Double-click **Product_ID** and **Profit** in the Selection pane to add both columns to the Select Data tab.
4. On the Select Data tab, click in the **Summary** column for the **Profit** column. Select the **SUM** statistic from the drop-down list. Notice that the **Automatically select groups** check box is selected.

   You can double-click **SUM_of_Profit** to modify the properties, including column name, label, and format.



5. Click the **Filter Data** tab. Drag **SUM_of_Profit** from the Selection pane to the Filter the summarized data pane.
6. In the New Filter Wizard, verify that the column name is **SUM_of_Profit**. Select **Greater than** in the **Operator** field. Enter **500** in the **Value** field and click **Finish**.

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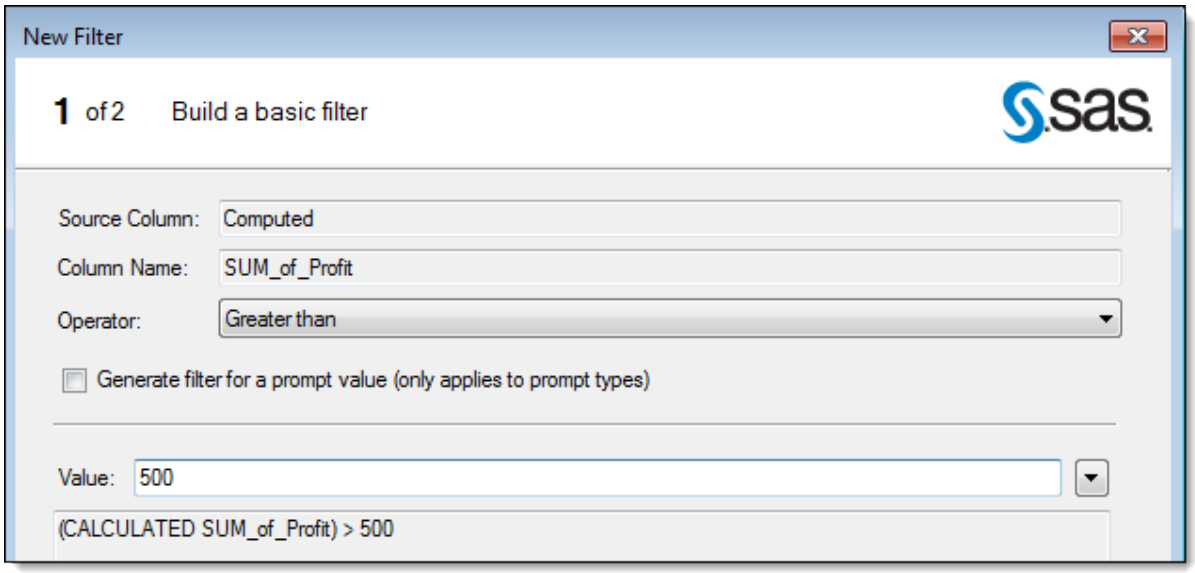

7. Click the **Sort Data** tab. Drag and drop **SUM_of_Profit** onto the tab area and change the sort direction to **Descending**.
8. Click **Run** and verify the results. Save the project.

   Partial Results (8 of 19 rows)

	Product_ID	SUM_of_Profit
1	230100700009	$1,660.65
2	230100700008	$1,578.50
3	230100700011	$1,322.00
4	240200200013	$1,139.40
5	240200100173	$946.40
6	240400200097	$877.80
7	240300300090	$850.65
8	240300300070	$825.45

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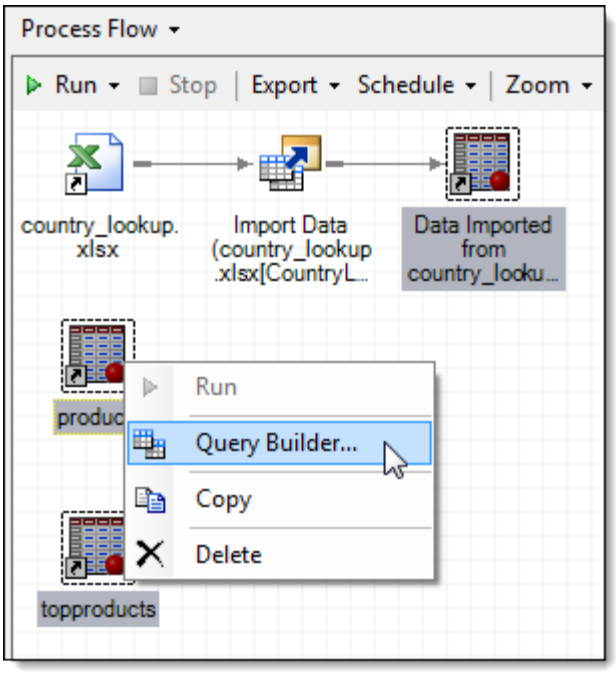
Joining Tables

In a previous query, products with total profits exceeding $500 were identified. Analysts asked for more details about these top products, including the product category, the product, supplier, and country name. To include the necessary columns, use the Query Builder to join the **TopProducts** SAS table with the **products** SAS table and the **country_lookup** Excel spreadsheet.

1. In the process flow, add the **TopProducts** and **Products** SAS data sets and the **country_lookup** Excel file to the project.
2. The Import Data task opens to import the **country_lookup** Excel file. Name the imported table **country_lookup** and select the **First row of range contains field names** check box. Do not make any changes to the field attributes.
3. In the process flow, hold down the Ctrl key and select **TopProducts**, **Products**, and **Data Imported from country_lookup**. Right-click one of the three tables and select **Query Builder**.

   If you are using an older version of Enterprise Guide, start building the query and use only one of the three tables. Then select **Add Tables** to add the other two data sets.



4. A message appears and indicates that Enterprise Guide was unable to find matching columns on which to join in all three tables. Click **OK**.
5. The Tables and Joins window automatically appears. To join the **products** and **country_lookup** tables, first select **Supplier_Country** in the **products** table. Select **Supplier_Country** a second time and drag it to connect it to **Country_Key** in the **country_lookup** table.

   You can also right-click **Supplier_Country** and select **Join with** ⇨ <***choosetable***> ⇨ **Country_Key**.

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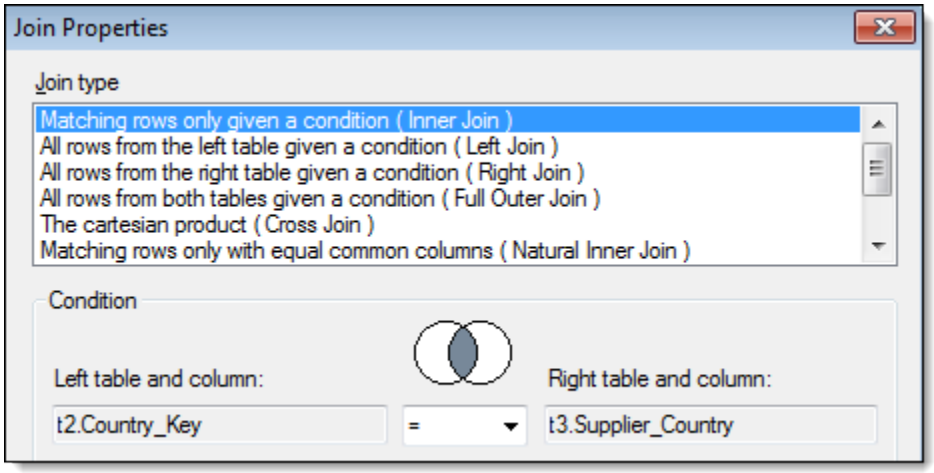
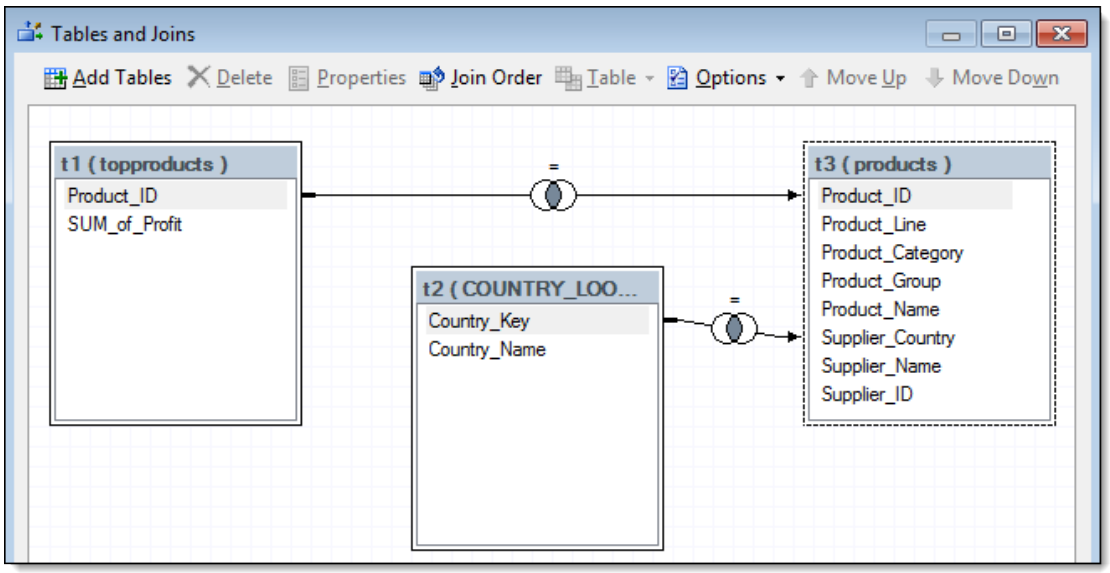
6. The Join Properties dialog box automatically appears. Verify that the join type is **Matching rows only given a condition (Inner Join)** and that the condition is **t2.Country_Key = t3.Supplier_Country**.

   The table alias names (**t1**, **t2**, or **t3**) might differ depending on the order in which the data sets were added to the query.

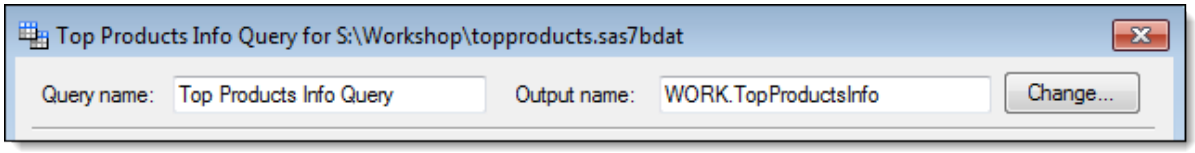


7. Click **OK** and verify that the three tables are linked as displayed below. Click **Close** to return to the Select Data tab.

   The order of the tables might differ in the Tables and Joins window.



8. Name the query **Top Products Info Query** and the output table **TopProductsInfo**.



9. Double-click the following columns to add each to the Select Data tab: **Product_ID**, **SUM_of_Profit**, **Product_Category**, **Product_Name**, **Supplier_Name**, and **Country_Name**.

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10. Click the **Sort Data** tab and sort the data by **SUM_of_Profit** in descending order.
11. Click **Run** to execute the query. Verify the results and save the project.

    Partial Results (8 of 19 rows)

	Product_ID	SUM_of_Profit	Product_Category	Product_Name	Supplier_Name	Country_Name
1	230100700009	$1,660.65	Outdoors	Family Holiday 6	Petterson AB	Sweden
2	230100700008	$1,578.50	Outdoors	Family Holiday 4	Petterson AB	Sweden
3	230100700011	$1,322.00	Outdoors	Hurricane 4	Petterson AB	Sweden
4	240200200013	$1,139.40	Golf	Master Golf Rain Suit	Mike Schaeffer Inc	United States
5	240200100173	$946.40	Golf	Proplay Executive Bi-Metal Graphite	Van Dammeren International	Netherlands
6	240400200097	$877.80	Racket Sports	Smasher Tg 70 Tennis String Roll	British Sports Ltd	United Kingdom
7	240300300090	$850.65	Indoor Sports	Top R&D Long Jacket	Top Sports Inc	United States
8	240300300070	$825.45	Indoor Sports	Top Men's R&D Ultimate Jacket	Top Sports Inc	United States
9	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

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